Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/20/2016
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Validation of a Cephalad Fluid Shift Countermeasure		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	brandon.r.macias@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Johnson Space Center Cardiovascular and Vision Laboratory		
PI Address 2:	2101 NASA Parkway, HAC/B21N-1207		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	06/01/2015	**End Date:**	05/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Change in period of performance per NSBRI (formerly 5/1/15-4/30/17)--Ed., 7/7/15		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sargsyan, Ashot M.D. (Wyle Laboratories) Cole, Christine (Clemson University) Kesari, Santosh M.D., Ph.D. (John Wayne Cancer Institute) Stenger, Michael Ph.D. (Wyle Laboratories) Ebert, Douglas Ph.D. (Wyle Laboratories) Hargens, Alan Ph.D. (University of California, San Diego) Liu, John Ph.D. (University of California, San Diego) Lee, Stuart M.S. (Wyle Integrated Sciences and Engineering Group) Huang, Alex M.D., Ph.D. (University of California, Los Angeles)		
Grant/Contract No.:	NCC 9-58-SMST04202		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project aims to provide critical data required to objectively determine how an optimized thigh cuff could be incorporated into the NASA integrated physiological countermeasure suite. This project will determine if thigh cuffs used during simulated spaceflight impact intracranial pressure (ICP), ocular structure and function, and intraocular pressure (IOP) using state-of-the-art techniques. This study will determine the physiological responses of thigh cuff application and removal on ocular and cerebral variables (including invasive ICP) in a microgravity analog. Furthermore, this proposed study will determine tissue pressure distribution applied by thigh cuffs in order to improve comfort, mobility, and efficacy of the countermeasure. Our team has designed and produced thigh cuffs, which address problems previously identified with the Russian Braslet. Thigh cuffs were produced in multiple geometries for improved anthropometric design, with micro and macro adjustments, in addition to a series of functional enhancements. Functional enhancements include usage of lighter weight, more flexible materials in fewer layers; improved closures; and ability to focus compression on site specific regions of the thigh. Focus is on achieving enhanced performance through using well-characterized materials, which are preferably commercially available in quantity. During year one of this project our team has acquired the necessary testing equipment, implemented training, and established protocols to initiate human cuff experiments during simulated microgravity at our co-investigator test sites. We have successfully recruited normal healthy subjects for initial cuff experiments scheduled for May, 2016. Early in year two we will conduct skin contact pressure experiments to determine the optimal thigh cuff width to be used during simulated microgravity experiments. Our team will finalize recruitment of subjects for our invasive measures of intracranial pressure during simulated microgravity with thigh cuffs. We will implement and complete testing using the optimized thigh cuff to determine how this device impacts ocular, vascular, and intracranial pressure variables during simulated microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Management of ocular and intracranial fluid volumes may be important variables that contribute to vision health during space flight and of patients on Earth. If successful, this countermeasure may provide clinicians with a new tool to investigate intracranial hypertension and reduce swelling of the optic nerve head region.
Task Progress:	During year one of this project our team has acquired the necessary testing equipment, implemented training, and established protocols to initiate human cuff experiments during simulated microgravity at our co-investigator test sites. We have successfully recruited normal healthy subjects for initial cuff experiments scheduled for May, 2016. Early in year two we will conduct skin contact pressure experiments to determine the optimal thigh cuff width to be used during simulated microgravity experiments. Our team will finalize recruitment of subjects for our invasive measures of intracranial pressure during simulated microgravity with thigh cuffs. We will implement and complete testing using the optimized thigh cuff to determine how this device impacts ocular, vascular, and intracranial pressure variables during simulated microgravity.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Articles in Peer-reviewed Journals	Berg-Johansen B, Liebenberg EC, Li A, Macias BR, Hargens AR, Lotz JC. "Spaceflight-induced bone loss alters failure mode and reduces bending strength in murine spinal segments." Journal of Orthopaedic Research. 2016 Jan;34(1):48-57. Epub 2015 Aug 31. http://dx.doi.org/10.1002/jor.23029 ; PubMed PMID: 26285046 , Jan-2016
Articles in Peer-reviewed Journals	Mao CP, Macias BR, Hargens AR. "Shoulder skin and muscle hemodynamics during backpack carriage." Appl Ergon. 2015 Nov;51:80-4. Epub 2015 May 19. http://dx.doi.org/10.1016/j.apergo.2015.04.006 ; PubMed PMID: 26154207 , Nov-2015
Articles in Peer-reviewed Journals	Macaulay TR, Macias BR, Lee SMC, Boda WL, Watenpaugh DE, Hargens AR. "Treadmill exercise within lower-body negative pressure attenuates simulated spaceflight-induced reductions of balance abilities in men but not women." npj Microgravity. 2016;2:16022. Published online: 30 June 2016. http://dx.doi.org/10.1038/npjmgrav.2016.22 , Jun-2016